

2.2 DYNAMICS QUESTIONS P2&P3

Compiled by: R. Muga

Q1: NOV 2007 P2

- 2 (a) Define *linear momentum*.

_____ [1]

- (b) A neutron moving with a velocity of 1.24×10^7 m/s collides head on with an alpha particle which is moving in the same direction with a velocity of 5.40×10^6 m/s.

- (i) If the collision was elastic, state and explain deductions one can make about momentum and velocity.

momentum _____

explanation _____

Q2: NOV 2009 P2

- 2 (a) Explain the difference between *elastic* and *inelastic* collision.

_____ [2]

- (b) A car of mass 1 tonne travelling at 30 ms^{-1} due East collides head-on with a lorry of mass 3 tonnes travelling at 20 ms^{-1} due West. Find

- (i) the velocity after collision if the car and the lorry move off together,

velocity = _____

- (ii) the change in kinetic energy.

Q3:JUN 2012 P2

- 1 (a) Define *velocity* and state its SI unit.

[2]

- (b) A volume of gas is enclosed in an insulated container by a frictionless piston. A molecule of mass, m , moving with speed, u , collides head-on with a stationary piston as in Fig.1.1.

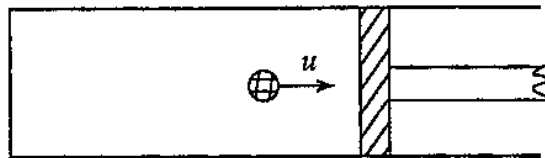


Fig.1.1

- (i) Write down an expression for the molecule's momentum change.

- (ii) State an assumption that you made.

[2]

- (c) The piston is moved outwards with a velocity, v , and collides with the gas molecule in (b). The molecule rebounds with a velocity, v_1 , as in Fig. 1.2.

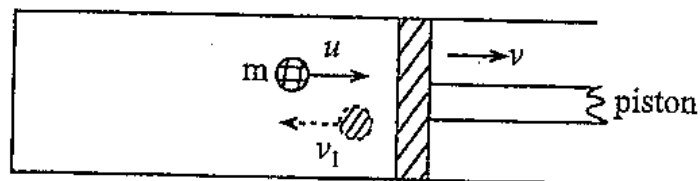


Fig.1.2

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- (i) Express v_1 in terms of u and v .

- (ii) Hence deduce the effect on the gas, of this movement by the piston.

[5]

- 2 (a) Define

- (i) *force*,

- (ii) *the newton*.

[5]

- (b) A body of mass 4.0 kg is pulled up a smooth plane inclined at 30° to the horizontal by a force of 40.0 N acting parallel to the plane.

- (i) Sketch a diagram showing all forces acting on the mass.

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(ii) Calculate

1. the resultant force, neglecting all frictional forces,

resultant force = _____

2. the acceleration of the body up the incline.

acceleration = _____

[5]

- (c) Suggest why, in practice, the value of acceleration is less than that in b (ii) 2.

[1]

Q4:JUN 2013 P2

- 2 (b) Distinguish between *elastic collision* and *inelastic collision*.

[2]

- (c) A mass, m , travelling with velocity, u , collides with a stationary mass, M . After the collision the two masses move off with speed, v and V , respectively in the original direction of m .

- (i) Assuming the collision to be elastic, write down the equation for the conservation of

1. kinetic energy,

2. momentum.

- (ii) Hence show that $V = \frac{2mu}{M+m}$.

[4]

Q5:JUN 2014 P2

- 2 (a) Define *impulse*.

[1]

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- (b) A 2 kg body at rest is subjected to a 200 N force for 0.20 s followed by a 400 N force for 0.30 s. The forces are acting in the same direction.

Determine

- (i) the momentum of the body,

momentum = _____

- (ii) the final speed of the body.

final speed = _____

[4]

- (c) Explain why, when a tennis player strikes a ball, he 'follows through'.

[3]

Q6: NOV 2015 P2

2

- (a) Define *acceleration*.

[1]

- (b) A ball of mass 50.0 g is thrown vertically downwards with a velocity of 3.25 ms^{-1} . It hits the ground after 1.62 s.

- (i) Assuming air resistance to be negligible, calculate the

1. speed of the ball when it hits the ground,

- (ii) The ball makes contact with the ground for 15 ms and rebounds with an upward velocity of 15.2 ms^{-1} .

Calculate the

1. average force acting on the ball on impact with the ground,
2. maximum height the ball reaches after hitting the ground.

maximum height _____

[8]

Q7:JUN 2016 P2

- 2 (a) (i) State the principle of conservation of momentum.

- (ii) A car of mass 1200 kg travelling at 40 ms^{-1} rams into the back of an unloaded lorry of mass 3000 kg travelling at 25 ms^{-1} and they coalesce.

Calculate the velocity of the vehicles after collision.

velocity = _____

[4]

- (b) Explain how seat belts in vehicles reduce injuries.

[2]

Q8:NOV 2016 P2

- 2 (a) State the principle of conservation of momentum.

[2]

- (b) Fig. 2.1 shows a snooker ball of mass 0.40 kg travelling at 2.0 m/s colliding head on with another identical ball travelling at 1.0 m/s. After the collision, the first snooker ball bounces back at 0.50 m/s.

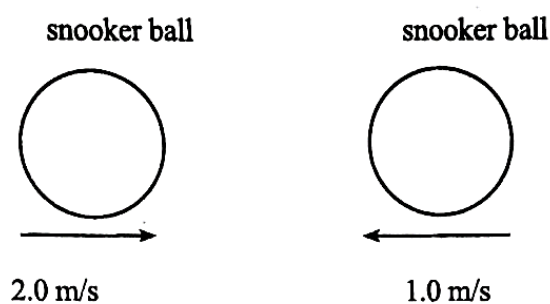


Fig. 2.1

- (i) Show that the second ball bounces back as well.
- (ii) Determine whether the collision is elastic or inelastic.

[5]

- (c) Explain why a child jumping off a wall should bend knees on landing.

[2]

Q9:JUN 2018 P2

- 2 (a) Define *linear momentum*

[1]

- (b) A trolley, x, of mass 3 kg is moving with a velocity of 5 m/s. The trolley collides head on with another trolley, y, of mass 2 kg moving in the opposite direction with a velocity of 6 m/s as shown in Fig.2.1.



Fig.2.1

After the collision, the two trolleys couple and move with a common velocity, v .

- (i) Calculate the total momentum before the collision.

total momentum before the collision = _____

- (ii) Determine the common velocity, v .

Q10: NOV 2003 P3

- 1 (a) (i) What is meant by *linear momentum*?
- (ii) State the principle of conservation of linear momentum.
- (iii) Explain how the principle is linked to Newton's third law. [6]
- (b) Two spheres A and B moving in the same directions as shown in Fig. 1.1 collide elastically. Sphere A has mass m and speed $4u$ and sphere B has mass $2m$ and speed u . The final speeds of A and B after collision are v_1 and v_2 respectively.

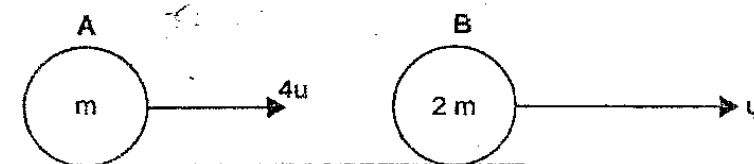


Fig. 1.1

- (i) What is meant by an *elastic* collision?
- (ii) Write down an equation for the conservation of momentum.
- (iii) Write down an equation for the conservation of energy.
- (iv) Deduce the value of v_2 in terms of u . State what happens to the motion of A after collision. [7]
- (c) In another interaction, a plasticine sphere of mass 0.30 kg is dropped from a height of 5.00 m and sticks to the ground.
- (i) Deduce the momentum of the sphere just before impact.
- (ii) State and explain the transfer of energy and momentum after impact. [7]

Q11: NOV 2006 P3

- 1 (a) Define *linear momentum*. [1]
- (b) A neutron moving with velocity 3.96×10^7 m/s collides head on elastically with a helium atom moving with a velocity of 2.97×10^7 m/s in the opposite direction. (mass of helium = $4u$).
- (i) State and explain deductions one can make from this statement about momentum, energy and relative velocity.
- (ii) Calculate the velocities of the neutron and helium after the collision. [10]

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- (c) Fig. 1.1 shows the variation of force applied to a tennis ball by a racquet with time.

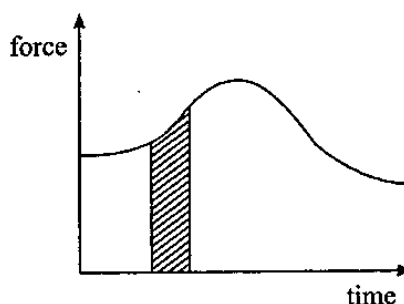


Fig. 1.1

- (i) State the quantity represented by the area under the graph.
- (ii) When a tennis player strikes a ball he/she 'follows through' in order to keep the racquet in contact with the ball for a long time. Explain.
- (iii) A tennis ball of mass 53.0 g initially at rest was acted upon by a force of 200 N for a time of 0.060 s. Calculate its velocity after impact.
- (iv) After the impact, total momentum is conserved. Comment on the validity of this statement.

[9]

Q12:JUN 2012 P3

- 1 (c) (i) State the *principle of conservation of linear momentum*.
- (ii) A particle O, of mass, U , travelling with a velocity of $3.0 \times 10^7 \text{ ms}^{-1}$, collides elastically with another particle, P, of mass $16U$, initially at rest. After collision particle O moves in a direction at 90° from its original path.

Using resolution of vectors and the principle of conservation of momentum, calculate the speed of P after collision.

[7]

Q13:NOV 2012 P2

- 2 (a) State the *principle of conservation of linear momentum*. [2]
- (b) A stationary ball of mass, $6.0 \times 10^{-2} \text{ kg}$ is hit horizontally with a tennis racket. The ball is in contact with the racket for 30 ms and leaves the racket with a speed of 27 ms^{-1} .
- (i) Calculate the
 1. change in momentum of the ball,
 2. average force which the racket exerts on the ball,
 3. horizontal distance travelled by the ball before it hits the ground if it leaves the racket at a vertical height of 2.5 m.

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- (ii) Explain what is meant by *an inelastic collision*.
- (iii) Suggest a reason why the collision between the ball and the racket is inelastic.

[8]

Q14:JUN 2013 P3

6

- 2 (a) Define the term *elastic collision*. [2]
- (b) A motorist travelling at 15 m/s on a busy day approaches a traffic light which turns red when he is 20 m away from the stop line.

Given that his reaction time, to start slowing down, is 0.3 seconds and he slows down at 4.5 m/s^2 , determine the distance the car stops from the stop line. [6]
- (c) From your solution in (b) deduce whether an accident is likely to occur. [2]

Q15:JUN 2014 P3

- 3 (a) State the principle of conservation of momentum. [1]
- (b) In an elastic head-on collision, a ball of mass 500 g moving at 4.0 m/s collides with a stationary ball of mass 1.0 kg.
 - (i) Calculate the velocities of the two balls after the collision.
 - (ii) If the balls had stuck and moved off together as one body after colliding, explain how their **total** energy would be affected. [7]
- (c) State and explain any **one** application of the law of conservation of momentum. [2]

Q16:JUN 2015 P3

- 1 (b) (i) State Newton's third law of motion.
- (ii) A stone falls towards the earth but the earth does not move up towards the stone.

State, with reasons, whether Newton's third law is obeyed in this case.

[5]

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Q17: NOV 2015 P3

- 2 (a) Distinguish between a vector quantity and a scalar quantity. [2]
- (b) A car travelling at 20 ms^{-1} due East turns and continues at 15 ms^{-1} due South.
- (i) Draw a vector diagram to illustrate the
1. initial and the final velocity,
 2. change in velocity.
- (ii) Calculate the change in the
1. speed of the car,
 2. velocity of the car.

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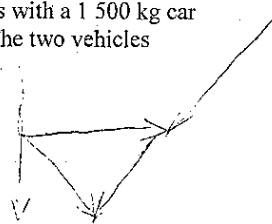
[6]

Q18: NOV 2016 P3

- 3 (a) (i) Define *impulse*.
- (ii) State the SI unit of impulse. [2]
- (b) A 7 500 kg lorry travelling at 5 m/s due east collides with a 1 500 kg car moving at 20 m/s in the direction South 60° West. The two vehicles coalesce.

Determine the

- (i) direction and
- (ii) speed
- of the wreckage. [6]



- (c) Fig. 3.1 shows some of the safety features in a car which include safety belt and head restraint. Explain how the head restraint prevents the driver from injury if the car is hit from behind. [2]

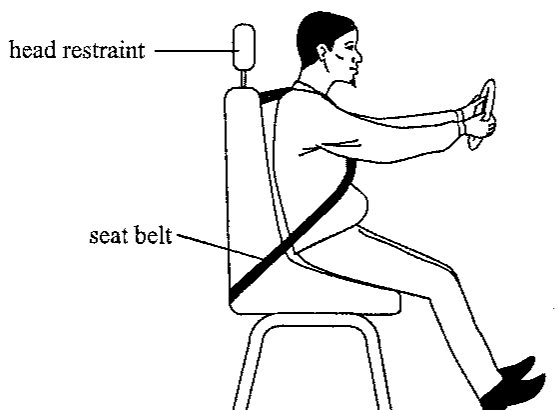


Fig. 3.1

Q19: NOV 2008

- 4 (a) State Newton's second and third laws of motion. [2]

- (b) A constant force of 9.5 N acts on a sphere A of mass 230 g lying on a smooth horizontal surface as shown in Fig. 4.1.

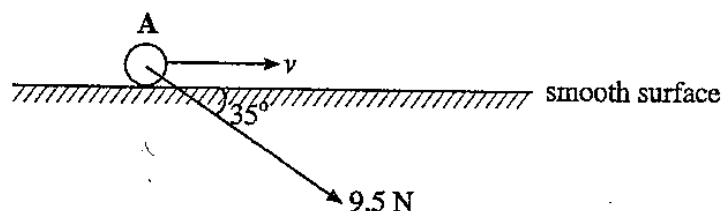


Fig. 4.1

The sphere A starts from rest and after 5.00 s the force is removed. Sphere A collides elastically with sphere B of mass 250 g initially at rest.

- (i) Calculate
1. force exerted on sphere A by the surface,
 2. the change in momentum of sphere A before collision,
 3. the velocity of sphere A just before collision.
- (ii) Write an equation relating the relative velocities in the collision. Hence or otherwise deduce the velocity of sphere A after the collision. [8]

Q20: NOV 2009

- 4 (a) Define *linear momentum* and *impulse*. [2]

- (b) Use Newton's third law to derive the principle of conservation of momentum. [4]

- (c) Two motor vehicles A and B are moving along a horizontal straight road in opposite directions. Vehicle A of mass 3 500 kg is moving with a speed of 14 m/s and vehicle B, mass 2 000 kg is moving with a speed 20 m/s. They collide and move together.

Calculate the change in kinetic energy. [4]

Q21: JUN 2013 P3

- 1 (b) Fig. 1.1 shows a textbook resting on a table.

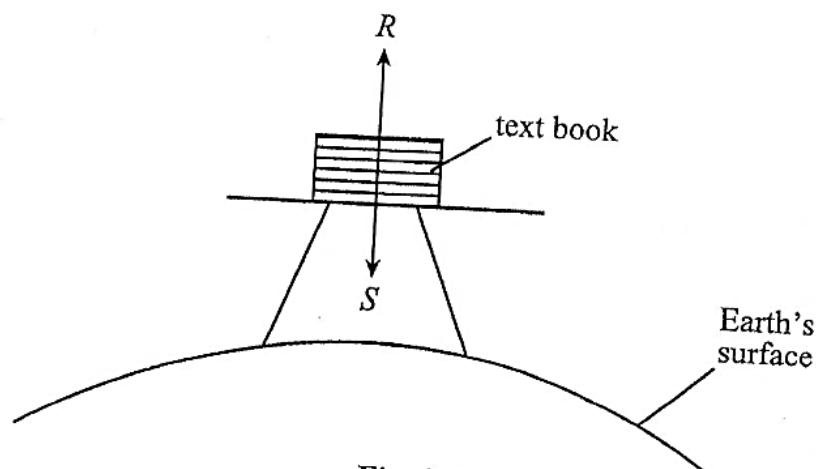


Fig. 1.1

R and S are two forces acting on the text book.

- (i)
 1. From Newton's third law of motion, state the body on which the force that pairs with S acts.
 2. State **one** way in which S and the force which pairs with it are similar.
- (ii) If the mass of the text book is 700 g, calculate S .
- (iii) With reference to Newton's second law of motion and your answer in (ii), comment on the effect of the force which pairs with S on the body on which it acts.

[5]

Q22: JUN 2014 P3

- 1 (b)
 - (i) State Newton's second law of motion.
 - (ii) Hence deduce how **The newton** is defined from this law.
 - (iii) A rocket is propelled by the emission of hot gases from the combustion of fuel. *The hot gases and the rocket gain momentum during the firing of the rocket.*

Discuss the statement in italics in relation to the principle of conservation of momentum.

[6]

Q23:JUN 2019 P3

- 1 (c) An object is sliding on a horizontal line XYZ where $XY = 24 \text{ m}$ and $XZ = 30 \text{ m}$. The object experiences a constant resistive force of 0.3 N . The speed of the object at X is 20 ms^{-1} and at Y is 16 ms^{-1} .

Calculate the

- (i) acceleration of the object,
- (ii) speed of the object at Z,
- (iii) weight of the object.

[4]

Q24:JUN 2017 P3

- 3 (a) State *the principle of conservation of momentum*. [2]
- (b) A boat of mass $3\,000 \text{ kg}$ moving at 5 m/s due north strikes a floating wooden log of mass $1\,000 \text{ kg}$ drifting due east with a velocity of 0.5 m/s . The boat and the wooden log remain locked together after collision.

Find their

- (i) direction of movement after collision,
- (ii) velocity after collision.

[6]

- (c) State and explain whether the momentum of the boat is conserved after collision. [2]